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Empirical evidence on disclosing cyber breaches in an 8-K report: Initial exploratory evidence

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ABSTRACT

The primary objective of this exploratory study is to assess whether 8-K disclosure, compared to alternative disclosure, explains the inconsistent results in prior studies examining the impact of cyber breaches on stock prices. The major finding of this study is that the negative magnitude of the stock price reaction and the duration of the reaction are substantially different for firms disclosing their cyber breaches via an 8-K compared to firms disclosing their breaches via other sources. Our results provide support for the new cybersecurity disclosure rules issued by the Securities and Exchange Commission (2023).

1. Introduction

Numerous studies assessed the impact of cyber breaches (i.e., cyber incidents) on stock prices of publicly traded corporations. A review of 45 studies concluded that most (75.6%) report evidence of a significantly negative stock price effect for firms that report a cyber breach (Spanos and Angelis, 2016). However, other studies conclude that the cyber breach stock price impact is not significantly negative or that the effect is mixed (Campbell et al., 2003; Hovav and D'arcy, 2003; Kannan et al., 2007; Gordon et al., 2011; Hilary et al., 2016; Richardson et al., 2019). Although most studies focused on a short event window effect (one to three days before and after the breach), several studies examined the stock market effects of breaches over a post-announcement period (i.e., weeks, months, or years) and these latter results have also been inconsistent (Morse et al., 2011; Hilary et al., 2016; Richardson et al., 2019; Bischoff, 2021).

Two important issues explaining the above noted inconsistencies are cyber breach *materiality* (i.e., a major breach that could affect shareholders' decisions about a company) and cyber breach disclosure *timeliness* (Amir et al., 2018; Foerderer and Schuetz, 2022). Disclosure in an 8-K addresses both issues. Thus, this study's primary objective is to provide exploratory evidence on whether 8-K breach disclosure, compared to alternative disclosure (e.g., 10-K, 10-Q, media, etc.), explains the inconsistent results concerning cyber breaches' impact on stock prices.

In 2011, the U.S. Securities and Exchange Commission (SEC) published "CF Disclosure Guidance: Topic No. 2," (hereafter, the 2011 Disclosure Guidance), that indicates investors have a right to know about a registrant's material cyber breaches (SEC, 2011). In SEC (2018), the Commissioners of the SEC revisited cybersecurity disclosure issues in their "Commission Statement and Guidance on Public

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Company Cybersecurity Disclosures" (hereafter, the 2018 Commission Statement). Both documents emphasize the importance of timely cyber breach disclosure in an 8-K. The SEC's (2023) new cybersecurity disclosure rules also emphasize the importance of timely disclosure of material cyber breaches in an 8-K.[1]

A large body of literature discusses the pros and cons associated with 8-K disclosure of material events (McMullin et al., 2019; Ben-Rephael et al., 2022). As discussed in the next section, firms may resist filing an 8-K cyber breach disclosure due to the potential negative reaction by shareholders.[2] In contrast, investors might reward firms for timely disclosures of major cyber breaches via an 8-K.

Using a sample of 670 public firm cyber breaches, we show that a significantly negative stock market reaction occurs around cyber breach announcements and this negative reaction persists, but is not significant, post-announcement. However, our results show that the negative reaction around a breach disclosure is statistically significant for firms with an 8-K breach disclosure, but statistically insignificant for firms with alternative breach disclosure methods. Furthermore, the post-announcement stock market reaction to 8-K breach disclosure firms is positive and statistically significant, but negative and statistically significant for firms with alternative breach disclosures. However, we cannot exclude that our results are at least partially due the nature of cyber breaches and not only to the channel used for disclosure.

To the best of our knowledge, this is the first study to examine the impact of 8-K cyber breach disclosures and it provides initial exploratory evidence on the potential effects of the new SEC 2023) rule about cyber breach disclosures in 8-K reports. As such, this study could be a useful baseline for future studies on the effects of the new SEC disclosure rule.

2. Literature review and hypotheses development

2.1. Background

The growing number and severity of cyber breaches highlights the importance of cybersecurity in an interconnected digital environment (Hill and Swinhoe, 2021). The interaction among financial disclosure, cybersecurity, and public policy is also an issue of growing importance (Ettredge and Richardson, 2003; Gordon et al., 2006; Yen et al., 2018; Wang et al., 2013; Ettredge et al., 2018; Haapamäki and Sihvonen, 2019; Janvrin and Wang, 2021; Calderon and Gao, 2022). Consequently, the SEC published its 2011 Disclosure Guidance, addressing investors' rights to information about registrant's material cyber breaches and cyber risks. Unfortunately, the 2011 Disclosure Guidance did not result in SEC registrants disclosing material cyber breaches in a timely manner (Amir et al., 2018). However, it did increase the number of registrants that voluntarily disclosed cyber incident and risk information in their SEC filings (Audit Analytics, 2016).

The concern over the transparent and timely disclosure of material cyber breaches reached a high in 2017, due in part to the Equifax cyber breach. Thus, the SEC revisited the issues of timely material cyber breach disclosure and insider trading on cyber breach information. This resulted in the 2018 Commission Statement, which elevated the visibility and importance of cybersecurity disclosure. The Commission Statement prominently discussed the importance of timely material cyber breach disclosure via an 8-K:

> The Commission encourages companies to continue to use Form 8-K … to disclose material information promptly, including disclosure pertaining to cybersecurity matters (p.10).

The 2011 Disclosure Guidance also mentioned the importance of using an 8-K for reporting cyber breaches. This recommendation, however, was relegated to an endnote.

The 2018 Commission Statement also provides additional clarity on cyber breach materiality:

> …The Commission considers omitted information to be *material* if there is a substantial likelihood that a reasonable investor would consider the information important in making an investment decision or that disclosure of the omitted information would have been viewed by the reasonable investor as having significantly altered the total mix of information available (p. 10).

The SEC's 2023 cybersecurity disclosure rules add Item 1.05 to Form 8-K, requiring registrants to disclose material cybersecurity incident information within four business days after identification.[3]

The benefits from filing an 8-K (i.e., the "current report") for timely disclosure of material events are well known. In 2004, the SEC adopted the "four business days" rule as the 8-K filing deadline (SEC, 2004). McMullin et al. (2019), for example, provide evidence that the SEC's 2004 decision to move the filing of an 8-K to within "four business days" of an event improved price formation. Ben-Rephael et al. (2022) found that institutional investors and media attention to Bloomberg terminals is significantly higher on 8-K filing dates.

2.2. Material and timely disclosure

An unambiguous definition of what constitutes a material cyber breach does not exist (Patterson and Smith, 2003). In fact, the 2018 Commission Statement notes that "…no 'single fact or occurrence' is determinative as to materiality, which requires an inherently fact-specific inquiry." Nevertheless, 8-K cyber breach disclosure provides *prima facie* evidence that the firm considers the event material in

[1] The new rules (SEC, 2023) went into effect as of December 15, 2023.

[2] The findings that reporting cyber breaches have a negative stock price effect have been relatively consistent, although the statistical significance of cyber breaches has been inconsistent.

[3] The four business days rule did not apply to foreign registrants prior to the SEC (2023) new rules.

nature (i.e., only major events that could affect investors' decisions are required to be reported in an 8-K). Thus, it seems reasonable to expect a significant and negative firm stock price reaction to an 8-K cyber breach disclosure. However, firms withhold or delay disclosing bad news due to *proprietary costs* (Verrecchia, 1983; Chambers and Penman, 1984; Dye, 1985; Amir and Ziv, 1997; Doyle and Magilke, 2009; Kothari et al., 2009; Amir et al., 2018). Amir et al. (2018) provide evidence supporting this view with severe cyber breaches. Similarly, Foerderer and Schuetz (2022) find that firms strategically time disclosure for severe breaches.

Bad news avoidance is especially true when the news cannot be observed by outside sources, which is common for cyber breaches (Amir et al., (2018). Furthermore, since cyber breach disclosure could give hackers information to facilitate additional cyber-attacks, additional incentives to minimize disclosure of cyber breach information exist. The SEC recognized the legitimacy of this incentive (e.g., 2018 Commission Statement, p. 11).

A counter argument to withholding, delaying, or strategically timing voluntary cyber breach disclosure is that firms may pre-emptively disclose bad news to minimize legal liability and/or potential reputation loss resulting from withholding bad news (Skinner, 1994, 1997; Wynn, 2008). The importance of cyber breach litigation risks was noted in the Governmental Accountability Office report (GAO, 2021). Firms might also prefer to voluntarily disclose breach information to signal to the stock market that the firm is pro-actively addressing cyber security issues (Gordon et al., 2010; Barry et al., 2022).

2.3. Hypotheses

Many factors affect the magnitude and duration of the cyber breach impact on stock prices. Two factors that seem especially relevant to this impact are cyber breach *materiality* and cyber breach disclosure *timeliness* (Amir et al., 2018; Foerderer and Schuetz, 2022; SEC, 2023). Unfortunately, differentiating between a material and non-material cyber breach is often ambiguous and cyber breach disclosure timeliness is typically driven by a variety of factors. However, cyber breaches reported in an 8-K provide a unique subset of breaches: firms are only required to file an 8-K, within four business days, for *major* breaches that could affect investors' decisions.

Firms face a dilemma because 8-K cyber breach disclosure could generate a larger negative stock price reaction surrounding the breach disclosure compared to withholding and disclosing via an alternative source. In contrast, due to the reputation effect, legal liability issues, and signaling potential discussed above, stock market recovery post-breach announcement may be faster when firms disclose breaches via an 8-K. In essence, the market could reward firms for the transparency and timeliness associated with 8-K cyber breach disclosures. By comparing the impact of 8-K cyber breach disclosure to alternative disclosure, we shed light on the above dilemma.

We establish a baseline for the effect of breach disclosure over a short event period (i.e., −1,1) and post-announcement period (i.e., 2, 20) in our analysis. This examination was conducted based on the following null hypotheses.

H_{1a}: Cyber breaches have no significant effect on short event window firm stock returns.
H_{1b}: Cyber breaches have no significant effect on post-announcement firm stock returns.

After establishing a baseline, we split the sample of firms with cybersecurity breaches into two groups: 1) 8-K cyber breach disclosers and 2) alternative source cyber breach disclosers. We test for differences between these groups based on the below null hypotheses.

H_{2a}: There is no significant difference in the impact on short event window stock returns of firms that disclose cyber breaches within an 8-K compared to firms that disclose cyber breaches through an alternative medium.
H_{2b}: There is no significant difference in the impact on post-announcement stock returns of firms that disclose cyber breaches within an 8-K compared to firms that disclose their cyber breaches through an alternative medium.

Table 1
Sample Selection and Sample Description.

Panel A: Sample Selection Process

	8-K filed	8-K not filed	Total
Raw sample of Cyber Breaches	90	767	857
Eliminate breaches from the same firm where another breach occurred within the last 240 calendar days	(10)	(110)	(120)
Eliminate breaches with missing stock return data	(7)	(60)	(67)
Breaches in final sample	73	597	670

Panel B: Cyber breach distribution in pre- and post- 2018 SEC Commission Statement

	8-K filed	8-K not filed	Total	Chi-square
Pre- 2018 SEC Commission Statement	30	315	345	3.5453
Post- 2018 SEC Commission Statement	43	282	325	(0.0597)
Whole sample	73	597	670	

Panel A presents the screening that we applied to obtain our final sample.
Panel B presents how the sample is distributed between the pre- and post- 2018 Commission Statement.
*Affected firms is in indication of the number of unique firms represented in the sample.

3. Data and methodology

3.1. Data sample

Our sample is derived from the Audit Analytics Cybersecurity Dataset and includes cyber breaches from March 2005 through February 2021. Table 1, Panel A, presents our sample selection process. The sample information included the cyber breach disclosure date, disclosure source, and firm characteristics. Our initial sample included 857 cyber breaches with 90 8-K disclosures and 767 alternative disclosures. We removed breaches occurring within 240 days of another breach within the same firm. After excluding events for firms without available stock return data, our final sample includes 670 observations, with 73 8-K disclosures and 597 alternative disclosures.

The subsample of 73 8-K reports includes 45 filed before or on the same day the breach was announced in another source. The 28 remaining 8-K reports were filed on breaches that were reported in another source prior to the 8-K filing. Of these 28 breaches, 17 breaches were reported via an 8-K within four business days of the announcement in another source, while 11 were reported in an 8-K more than four business days after disclosure elsewhere. Given the filing differences, we perform our analyses using both the sample of 73 breaches and a reduced sample of 62 breaches (i.e., 45 breaches plus the 17 discussed above). Table 1, Panel B, indicates that the frequency of 8-K disclosure has increased in the period following the 2018 Commission Statement release.

3.2. Event study methodology

To test the hypotheses, our first step examines whether the returns around our event day differ significantly from a normal return (Mackinlay, 1997). The event date is the cyber breach disclosure date.[4] We calculated the abnormal return for firm i at event date t using Equation (1):

$$AR_{it} = R_{it} - E(R_{it}|X_t), \tag{1}$$

where AR_{it} = abnormal return for firm i at event date t,

R_{it} = actual return for firm i at event date t,

$E(R_{it}|X_t)$ = expected return for firm i at event date t, estimated from 165 days before the disclosure date to 45 days before the disclosure date using the Fama-French 3-factor model based on the equal-weighted market index.[5]

Fig. 1 presents two graphs of the average abnormal returns for the days surrounding the cyber breach disclosure. Panel A presents the graph for all 8-K breach disclosures and Panel B presents only 8-K breach disclosures filed within four business days of a previous disclosure. As shown, disclosure has the largest impact on the returns of 8-K filers on the actual filing date (day 0), signaling the market recognized the disclosure on that date.

Using Fig. 1, we defined our event window as the day before the disclosure date and the day after the disclosure date (−1, 1). All tests computed abnormal returns in this event window, utilizing the cumulative abnormal return (CAR) as represented in Equation (2):

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{it} \tag{2}$$

We examined the average CAR for the full sample in the short event window (−1,1) and post-announcement window (2,20). We further divided the observations into the subsamples of 8-K and non-8-K disclosures to explore differences in the market returns between these two subsamples.

The second step of our event study used OLS to cross-sectionally analyze whether 8-K disclosure impacts the abnormal return of firms, as reported in Equation (3):

$$CAR_i = \beta_0 + \beta_1 * Filing8K_i + \beta_2 * HighTech_i + \beta_3 * ROA_i + \beta_4 * MV_i + \beta_5 * Analysts_i + \varepsilon_i \tag{3}$$

where

CAR_i = cumulative abnormal return for firm i.

$Filing8K_i$ = dummy variable set equal to 1 if cyber breach was disclosed by firm i using form 8-K, 0 otherwise.

$HighTech_i$ = dummy variable set equal to 1 if firm i belongs to one of the following SIC industry classes: drugs, R&D services, programming, computers, electronics, 0 otherwise.

ROA_i = net income divided by total assets in the year before the cyber breach for firm i.

MV_i = log of the market value of equity measured at the beginning of the year for firm i.

$Analysts_i$ = the number of analysts following firm i in the year of the cyber breach.

[4] For 8-K filing firms, 59 of the 73 8-K reports were solely dedicated to cyber breaches and no other significant events were identified in the remaining 14 8-K reports.

[5] We also conducted our event studies using the market model and the value-weighted market model, with similar results.

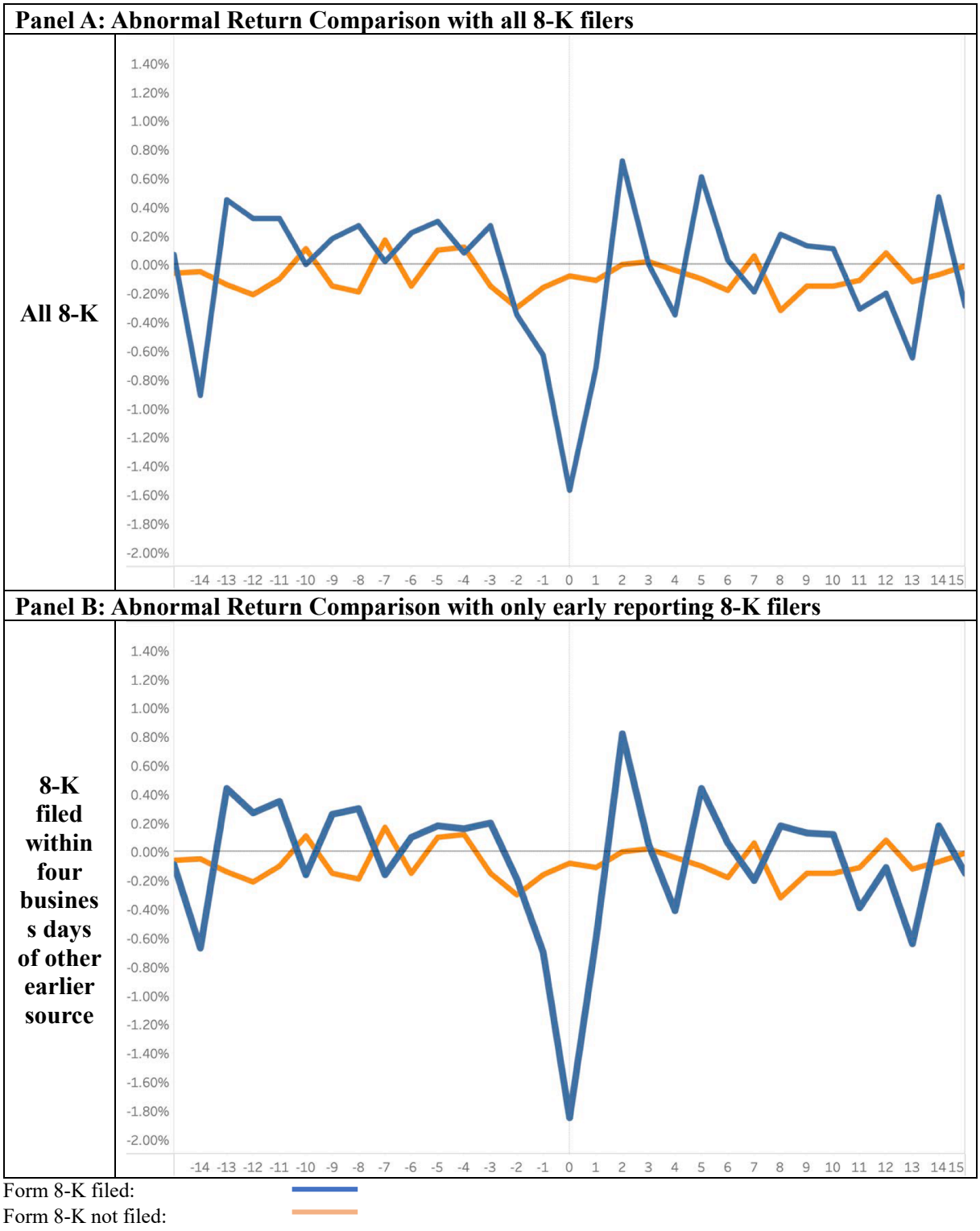

Fig. 1. Abnormal Return Comparison of 8-K vs. non-8-K Disclosures (−15, 15).

The control variables identified above are characteristics associated with disclosure in prior research (Skinner, 1994; Levitt, 1998; Amir et al., 2018). We also controlled for industry and year fixed effects.

4. Results

4.1. Descriptive statistics

Table 2, Panel A, presents statistics that provide insight into the size of firms that disclose cyber breaches. We tested for differences in means between our groups using both a pooled *t*-test, assuming equal variances, and the Satterthwaite *t*-test, assuming unequal variances. To test for differences between medians, we utilized the Wilcoxon-ranked-sum test. We observe that firms in the 8-K disclosure subsample are significantly smaller in terms of assets, sales, and market value compared to firms in the non-8-K disclosure subsample.

Table 2, Panel B, presents statistics associated with the variables identified in Equation (3). Using the same tests of medians and means as described above, we find firms that the 8-K disclosure subsample contains a greater percentage of high-tech firms and firms followed by significantly fewer analysts than the non-8-K disclosure subsample. Finally, the 8-k disclosure subsample is characterized by lower CAR during the short event window and a significantly higher CAR during the post-announcement window, compared to the non-8-K disclosure subsample. This evidence goes against our second hypothesis.

4.2. Event study results

Table 3 presents the results from the first step in our event study analysis. For the full, Panel A shows that the average *CAR* over the short event window (−1,1) is significantly negative (−0.63 %) at the 5 % level, leading us to reject H_{1a} that cyber breaches have no significant effect on the short event window stock returns. The average *CAR* over the post-announcement window (2, 20) is negative but insignificant, therefore we cannot reject H_{1b} that cyber breaches have no significant effect on post-announcement stock returns. After grouping the observations into 8-K disclosures vs. non-8-K disclosures, we find that the average *CAR* over the short event window (−1, 1) for 8-K disclosure is negative (*CAR* of −2.91 %) and significant at the 1 % level. In the post-announcement period (2,20), 8-K disclosures have a positive *CAR* (2.49 %) that is statistically significant at the 5 % level. In contrast, while non-8-K disclosures do not have a significant *CAR* in the short event window (−1,1), non-8-K disclosures are associated with a negative *CAR* of −1.04 % in the period following the disclosures (2,20) that is significant at the 5 % level. Furthermore, when we examined only 8-K disclosures occurring within four business days of the earliest breach disclosure, the 3-day stock reaction to the breach remains negative and increases in significance. Moreover, the CARs over the post-announcement period (2,20) remain positive and are statistically significant.

Table 2
Descriptive Statistics.

	Panel A: Descriptive statistics related to the first step in our event study								
	8-K filed			**8-K not filed**			**Comparison**		
	N	Mean	Median	N	Mean	Median	*t*-test Pooled	*t*-test Satterthwaite	Wilcoxon rank sum Z
Assets	73	45,731.32	2,890.49	593	77,379.94	9,681.00	−0.97	−0.90	−4.868***
Sales	73	8,494.49	1,948.90	593	29,468.80	5,591.30	−2.64***	−6.07***	−4.144***
Market Value	70	13,178.87	3,408.46	523	50,889.64	8,422.07	−2.31**	−5.35***	−3.830***
	Panel B: Descriptive statistics related to the second step in our event study								
	8-K filed			**8-K not filed**			**Comparison**		
	N	Mean	Median	N	Mean	Median	*t*-test Pooled	*t*-test Satterthwaite	Wilcoxon rank sum Z
HighTech	61	0.2787	0.0000	460	0.1739	0.0000	1.98**	1.73*	1.973**
ROA	61	0.0222	0.0380	460	0.0276	0.0405	−0.13	−0.25	−1.076
MV	61	8.1856	8.2140	460	9.3645	9.3680	−4.49***	−5.22***	−4.717***
Analysts	61	14.8853	12.0000	460	18.6761	17.5000	−2.25**	−2.74***	−1.962**
CAR(−1,1)	61	−0.0226	−0.0054	460	−0.0002	−0.0021	−3.17***	−2.40**	−1.622
CAR(2,20)	61	0.0292	0.0278	460	−0.0130	−0.0030	2.83***	2.63**	2.946***

Panel A presents descriptive statistics related to the size of firms under investigation and directly relates to our first step in our event study analysis. Panel B presents the descriptive statistics related to the disclosure characteristics used within the second step of our event study analysis.
Variables are described in section 3.
We test for differences between means using the Pooled (equal variances assumed) and Satterthwaite (unequal variances assumed) methods. We utilize the Wilcoxon-rank-sum method to test for differences between medians.
* Statistically significant at the 10 % level.
** Statistically significant at the 5 % level.
*** Statistically significant at the 1 % level.

Table 3
Cumulative Abnormal Return (*CAR*) Results for short event window and post-announcement window.

Panel A: Mean *CAR* over test windows

	3-day short event window (−1, 1)				19-day post-announcement window (2, 20)			
	Obs.	Median *CAR*	Mean *CAR*	Rank Test Z	Obs.	Median *CAR*	Mean *CAR*	Rank Test Z
All breaches	670	−0.24 %	−0.63 %**	−1.753**	669	−0.18 %	−0.66 %	−0.591
Breaches disclosed in 8-K report	73	−0.54 %	−2.91 %***	−2.254**	73	2.33 %	2.49 %**	1.964**
Breaches disclosed in 8-K report within four business days of other earliest source	62	−0.63 %	−3.16 %***	−2.436***	62	2.55 %	2.38 %**	2.050**
Breaches not disclosed in 8-K reports	597	−0.23 %	−0.35 %	−0.996	596	−0.38 %	−1.04 %**	−1.352*

Panel B: Difference in Mean *CARs* between 8-K filed and 8-K not filed

	3-day short event window (−1, 1)	19-day post-announcement window breach (2, 20)
Difference in mean *CARs*	−2.56%	3.54%
Pooled *t*-test	−3.07***	2.36**
Satterthwaite *t*-test	−2.26**	2.11**
Wilcoxon rank sum Z	−1.805*	2.475**

Panel C: Difference in Mean *CARs* between 8-K filed within four business days of other earliest source and 8-K not filed

	3-day short event window (−1, 1)	19-day post-announcement window (2, 20)
Difference in mean *CARs*	−2.81%	3.42%
Pooled *t*-test	−3.11***	2.13**
Satterthwaite *t*-test	−2.17**	1.02*
Wilcoxon rank sum Z	−1.738*	2.436**

Panel A presents the mean *CAR* analysis for two time periods: (1) the date of disclosure plus the days before and after, (2) the 19-day post breach time period.
Panel B presents a test for the differences between the returns reported by all 8-K disclosures and non-8-K disclosures during our time periods.
Panel C presents a test for the differences between the returns reported by 8-K disclosures within four business days of other earliest source and non-8-K disclosure during our time periods.
We test differences between means using the Pooled (equal variances assumed) and Satterthwaite (unequal variances assumed) methods. We utilize the Wilcoxon-rank-sum method to test for differences between medians. * Statistically significant at the 10 % level, ** Statistically significant at the 5 % level, *** Statistically significant at the 1 % level.

Table 3, Panel B, shows the tests for the differences between returns for 8-K disclosures and non-8-K disclosures. The tests indicate that 8-K disclosure results in a *CAR* that is −2.56 % lower than non-8-K disclosure in the short event window (−1,1) and that difference is statistically significant (1 % level based on pooled t-tests). In the period after the disclosure (2,20), 8-K disclosure leads to an average *CAR* that is 3.54 % higher than non-8-K disclosure and this difference is statistically significant at the 5 % level. In Panel C we present comparison results for the subset of breaches where 8-K disclosures were filed within four business days of other earliest source and breaches where 8-K disclosures were not filed, which are like those in Panel B. Thus, we reject H_{2a} and H_{2b}.

Table 4 presents the second step of our event study analysis related to H_{2a} and H_{2b}. The OLS estimates of model (3) are reported for both the days around disclosure (−1,1) and post disclosure (2,20). Panel A presents the OLS results using the full 8-K sample. Panel B presents the results using only 8-K disclosures that occurred within four business days of the earliest breach disclosure. The results in both panels are similar. The coefficients indicate that 8-K disclosures are associated with a 1.99 % lower return (5 % significance level) compared to non-8-K disclosures around the short event window. Additionally, our results show that 8-K disclosures are associated with a return that is 4.03 % higher than non-8-K disclosures (1 % significance level) in the period following initial public disclosure. Overall, the results indicate that 8-K disclosure firms experience returns that are significantly different from the returns of alternative disclosure firms in both the period surrounding the disclosures and period following the disclosures. These results lead us to reject H_{2a} and H_{2b}.

4.3. Additional analyses

When a major breach is discovered, law enforcement often asks the firm to not immediately announce the breach to facilitate tracing the hacker source. Additionally, firms want to remediate vulnerabilities causing the breach prior to public disclosure. Thus, the disclosure date, not the discovery date, is the key variable in our study.

We calculated the number of days between the breach date and the disclosure date for 117 breaches in our final sample. Thirteen (104) of these breaches are filed via 8-K (alternative) reporting and have an average time between breach and disclosure of 36.62 (140.68) days. We also compared the number of days between the time of discovery and the public disclosure for 8-K disclosures and non-8-K disclosures. We further calculated the number of days between discovery and disclosure for 296 breaches. Among these, 37 (259) are disclosed via 8-K (alternative) reporting and have an average 26.68 (60.73) days between the time of discovery and the

Table 4
Regression Analysis of *CAR* Results.

Panel A: Impact of 8-K disclosure on returns using sample including all 8-K reports

Independent Variables	Dependent Variable			
	***CAR* over Short event window (−1,1)**		***CAR* over Post-announcement Window (2,20)**	
	Coefficient	P-value	Coefficient	P-value
Intercept	−0.0064	0.475	0.1857	0.000***
Filing8K	−0.0199	0.024**	0.0403	0.000***
HighTech	−0.0087	0.194	−0.0069	0.553
ROA	−0.0239	0.000***	0.0951	0.000***
MV	0.0003	0.701	0.0020	0.618
Analysts	−0.0000	0.986	−0.0001	0.847
Industry FE	Included		Included	
Year FE	Included		Included	
Adj. R-Squared	9.44 %		13.50 %	
No. of Obs	521		521	

Panel B: Impact of 8-K disclosure on returns using sample including 8-K reports filed within four business days of other earliest source

Independent Variables	Dependent Variable			
	***CAR* over Short event window (−1,1)**		***CAR* over Post-announcement Window (2,20)**	
	Coefficient	P-value	Coefficient	P-value
Intercept	−0.0050	0.556	0.1812	0.000***
Filing8K	−0.0198	0.061*	0.0476	0.000***
HighTech	−0.0089	0.199	−0.0063	0.597
ROA	−0.0246	0.000***	0.0950	0.000***
MV	0.0002	0.865	0.0025	0.543
Analysts	−0.0000	0.982	−0.0002	0.807
Industry FE	Included		Included	
Year FE	Included		Included	
Adj. R-Squared	9.59 %		13.79 %	
No. of Obs	511		511	

Table 4 presents the OLS regression estimates of Equation (3): $CAR_i = \beta_0 + \beta_1{*}Filing8K_i + \beta_2{*}HighTech_i + \beta_3{*}ROA_i + \beta_4{*}MV_i + \beta_5{*}Analysts_i + \varepsilon_i$, for our entire sample of cyber breach disclosers. Variables are described in section 3.
* Statistically significant at the 10 % level
** Statistically significant at the 5 % level
*** Statistically significant at the 1 % level

public disclosure.

One important breach characteristic is whether a breach compromises financial versus non-financial data. We found that 39.7 % of 8-K reports are associated with financial data breaches as compared to 28.1 % of non-8-K reports. This difference is significant at the 5 % level.

5. Concluding remarks

Cyber breach disclosure in an 8-K provides *prima facie* evidence that a firm considers the breach to be a *material* event that should be disclosed in a *timely* manner. By comparing a set of 8-K cyber breach disclosures to alternatively disclosed cyber breaches, the current study finds that the negative stock reaction and the duration of the negative reaction are substantially different for 8-K disclosed breaches compared to alternatively disclosed breaches. These findings provide initial exploratory evidence on the expected effects of the SEC's new cybersecurity rules (2023).

As with all empirical studies, the one reported in this paper has limitations. The most important limitation is the small size of the sample of 8-K filings. Consequently, beyond being able to identify the fact that such breaches are associated with financial information, we could not provide a detailed understanding of the characteristics of the breaches filed in an 8-K nor could we identify specific firm characteristics that affect the stock market reactions to cyber breaches. The limited sample size also prevented us from conducting a detailed explanation concerning the 8-K post announcement returns or developing a model to predict 8-K vs. non-8-K disclosures. Hence, we cannot completely exclude that the differential results obtained in the 8-K disclosures sample may be due also to some of the intrinsic differences in the cyber breaches disclosed or in the firms disclosing through the 8-K.

One could also challenge our argument that that our sample of cyber breaches filed in an 8-K were material breaches because they were identified by the firms as being major breaches (i.e., only major breaches are required to be reported in an 8-K). Even if all material breaches are reported in an 8-K, non-material breaches could also be reported in an 8-K. The new disclosures rules by the SEC

(2023) specifically address this latter concern in Item 1.05 on the 8-K form. In addition, the emphasis on filing material cyber breaches in an 8-K under the new SEC rules will likely result in a significant increase in the number of breaches reported in an 8-K. Thus, within a couple of years, the new SEC rules should make it possible to meaningfully address the limitations noted above.

The above limitations notwithstanding, we believe the findings from the current study provide valuable exploratory evidence on the importance of reporting material cyber breaches in an 8-K vs. some other disclosure source. As such, this study provides a good baseline for future empirical studies examining the SEC's new rules concerning the filing of material cyber breaches in an 8-K.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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